

DAFFODIL INSTITUTE OF INFORMATION TECHNOLOGY (DIIT)
BSc. (Hons) in CSE 1st Year 1st Semester 24th Batch
Electrical and Electronic Circuit
Course Code: CSE-510203
Midterm Examination Suggestion

Suggested Topics List (Theory & Math)

1. Basic Circuit Laws & Theorems

- ✓ **Ohm's Law** – Theory & numerical problems
- ✓ **Kirchhoff's Voltage Law (KVL)** – Theory & circuit analysis problems
- ✓ **Kirchhoff's Current Law (KCL)** – Theory & circuit analysis problems
- ✓ **Voltage Divider Rule (VDR)** – Theory & practical calculations
- ✓ **Current Divider Rule (CDR)** – Theory & practical calculations
- ✓ **Thevenin's Theorem** – Theory, steps, and problem-solving
- ✓ **Norton's Theorem** – Theory, steps, and problem-solving
- ✓ **Superposition Theorem** – Theory & example circuits
- ✓ **Maximum Power Transfer Theorem** – Theory & mathematical proof

2. Semiconductor Physics

✓ **Definitions:**

- Diffusion, Drift Velocity, Diffusion Current
- Depletion Layer, Diode, Biasing
 - ✓ **Energy Band Diagram:**
- Conductors, Semiconductors, Insulators
 - ✓ **PN Junction (with diagrams):**
- Formation, Depletion Layer
- Forward & Reverse Bias (with diagrams)

3. AC Waveform Characteristics

- ✓ **Average Value** – Definition & derivation
- ✓ **RMS Value** – Definition & derivation
- ✓ **Form Factor** – Definition & calculation
- ✓ **Peak Factor** – Definition & calculation

4. Semiconductor Materials

- ✓ **Definition & Classification**
- ✓ **Intrinsic vs. Extrinsic Semiconductors**
- ✓ **P-Type & N-Type Semiconductors**

5. Rectifiers & Applications

- ✓ **Rectifier Definition & Purpose**
- ✓ **Half-Wave Rectifier** – Working principle & waveform
- ✓ **Full-Wave Rectifier** – Center-Tap & Bridge Rectifier (with diagrams & math)
- ✓ **Efficiency & Ripple Factor Calculations**

Mathematical Problems & Solutions: Basic Circuit Laws & Theorems

Topic	Theory	Mathematical Problems
Ohm's Law	Defines the relationship between voltage, current, and resistance. Formula: $V=IR$ or $V = IR$.	Find current/voltage/resistance in simple circuits.
Kirchhoff's Voltage Law (KVL)	The sum of voltages around a closed loop equals zero.	Solve circuit loops for unknown voltages.
Kirchhoff's Current Law (KCL)	The sum of currents entering a node equals the sum leaving.	Find unknown currents at junctions.
Voltage Divider Rule (VDR)	Used to determine voltage across resistors in a series circuit.	Compute voltage drops in series circuits.
Current Divider Rule (CDR)	Determines current through a branch in parallel circuits.	Solve for current distribution in parallel circuits.
Thevenin's Theorem	Any linear circuit can be simplified to a voltage source and series resistance.	Find Thevenin equivalent resistance and voltage.
Norton's Theorem	Any linear circuit can be represented as a current source in parallel with resistance.	Convert circuits to Norton equivalent forms.
Superposition Theorem	In a linear circuit with multiple sources, the total response is the sum of individual responses.	Solve circuits by considering one source at a time.
Maximum Power Transfer Theorem	Maximum power is transferred when load resistance equals Thevenin resistance.	Find optimal load resistance for maximum power transfer.